

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/251998039>

Simple power analysis attacks using chosen message against ECC hardware implementations

Article · January 2011

CITATIONS

3

READS

259

5 authors, including:



[Guoqing Xu](#)

SIAT,CAS

175 PUBLICATIONS 3,332 CITATIONS

[SEE PROFILE](#)

Simple Power Analysis Attacks Using Chosen Message against ECC Hardware Implementations

Huiyun Li, keke Wu, Guoqing Xu, Hai Yuan
Shenzhen Institutes of Advanced Technology,
Chinese Academy of Sciences

The Chinese University of Hong Kong
Shenzhen 518055, P. R. China

Email: (hy.li, kk.wu, gq.xu, hai.yuan)@siat.ac.cn

Peng Luo

State Key Laboratory of Information Security
Institute of Software, Chinese Academy of Sciences
Graduate University of Chinese Academy of Sciences

Beijing 100049, P. R. China
Email: luopeng@is.iscas.ac.cn

Abstract

Chosen-message simple power analysis (SPA) attacks were powerful against public-key cryptosystem based on modular exponentiation, due to the special results of modular square and modular multiplication for input pair X and $-X$. However, the characteristics can not be applied to public-key cryptosystems based on scalar multiplications. This paper proposes novel chosen-message side-channel analysis attacks for public-key cryptosystems based on scalar multiplications, where special input point P is chosen close to X -axis to generate noticeable variations for point doubling and point addition. The proposed attack can be applied to all standard implementations of the binary algorithms, both for left-to-right and right-to-left methods. This chosen-message method can also circumvent typical countermeasures such as the double-and-add-always algorithm.

1 Introduction

Elliptic curve cryptosystems (ECC) were independently introduced by Miller [1] and Koblitz [2] in the 1980s, which attracted increasing attention in recent years due to the requirement of shorter private key length with the same security level, in comparison with other public-key cryptosystems such as RSA.

Although the cryptosystems are secure, information about secret data may leak through side-channels analysis attacks (e.g., timing, power consumption, or electromagnetic emission), based on the fact that side-channel information will depend on the data being manipulated. In the scope of public-key cryptosystems, the simple power analysis (SPA) [3] attack is a commonly adopted technique which involves examining the power consumed by a device running public-key cryptographic algorithms over time. Variations in power consumption occur as the device performs different operations. Squaring and multiplication operations in RSA implementations can often be distinguished, enabling an adversary to compute the secret key. Classic ECC binary algorithm is analogous to the “square-and-multiply” principle used in RSA. If the point addition and doubling operations in ECC implementations can be distinguished, an adversary is able to obtain the secret key.

However, most current implementations of RSA implementations use the same sequence of instructions to implement multiplication and squaring, and ECC implementations use the same sequence of instructions for point addition and doubling, it thus becomes difficult to distinguish the pattern differences for random input. In order to enhance secret information leakage on the power waveforms, chosen-message attacks for RSA using specific input data were proposed [4, 5, 6, 7]. Yen proposed the use of the input value of -1 for the chosen-message [7] and discussed the possibility of an attack to defeat the most popular SPA countermeasures using dummy multiplication. In addition

to -1 , input values of X and $-X$ were used. As a result, the secret key information was clearly revealed in all of the RSA implementations. The basic concept is to use a specific input data, -1 , to enhance the differences between the multiplication and squaring operations. For example, let $-1 \bmod N = (N-1)$ be the input. Then the outputs of the multiplication and squaring operations during exponentiation are $-1 \bmod N = (N-1)$ and $1 \bmod N$, respectively. According to this property, the multiplication and squaring operations during exponentiation can be classified into three types: (M) multiplication after squaring, (S1) squaring after multiplication, and (S2) squaring after squaring as follows:

$$(M) Z = 1 * (-1) \bmod N = -1 \bmod N \quad (1)$$

$$(S1) Z = (-1) * (-1) \bmod N = 1 \bmod N \quad (2)$$

$$(S2) Z = 1 * 1 \bmod N = 1 \bmod N. \quad (3)$$

In the binary algorithm, squaring S1 follows multiplication M, and squaring S2 follows squaring S2 or squaring S1. In other words, M is never followed by S2. Thus, the bit pattern of the secret exponent can be obtained if one of the three operations is distinct from the others. In practical implementations, M and S1 operations consume much larger amount of power than S2, thus the key bit sequence can be identified [4]. Figure 1 illustrates the differences between M, S1 and S2 operations in the form of side channel trace patterns.

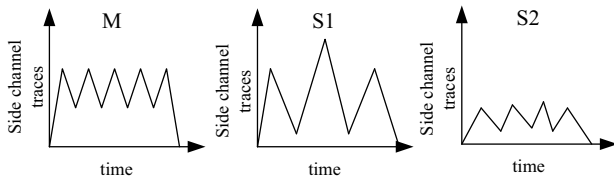


Figure 1. chosen-message against RSA

By confining the operation sequences as well as the data values, Yen's method can also defeat the "square-and-multiply always" algorithms. However, the authors in [4, 7] admitted that the chosen-message attack method can not extend to ECC-type of cryptosystems, due to the special characteristics of congruence in modular multiplication.

This paper will propose a chosen-message attack method against ECC, exploiting the special characteristics of the point at infinity, which causes variations in power consumption of point doubling and point addition. The rest of the paper is organized as follows. Section II introduces the fundamentals of ECC cryptosystems. In Section III, we describe the proposed chosen-message attack against ECC binary algorithm in details. In Section IV, we extend the attack scenario to the "double-and-add always" algorithm. We summarize our paper in Section V.

2 Introduction of ECC

An elliptic curve (EC) cryptosystem translates the existing discrete-logarithm problem into the context of EC. The discrete logarithm problem in a cyclic group G of order n with generator g refers to the problem of finding x given the element $y = g^x$ of G . The discrete logarithm problem over an EC is much harder than other groups such as the multiplicative group of a finite field. Let $E(F_p)$ be an elliptic curve defined over a finite field F_p . Let P be a point on $E(F_p)$. The prime number p , the equation of the elliptic curve F_p , and the point P and its order n , are the public domain parameters. A private key is an integer d that is selected uniformly at random from the interval $[1, n-1]$, and the corresponding public key is $Q = dP$. The problem of determining the private key d is thus the elliptic curve discrete logarithm problem (ECDLP).

2.1 Elliptic curve preliminaries

1) Definition

An elliptic curve (EC) is the set of points (x, y) which are solutions of a bivariate cubic equation over a finite field F . It can be expressed as the form of Weierstrass equation: $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$, where $a_i \in F$, defining an EC over F . If characteristic (char) $F \neq 2, 3$, Weierstrass equation can be transformed to: $y^2 = x^3 + ax + b$, with $a, b \in F$. The set of points on an EC, together with a special point O called the point at infinity can be equipped with an Abelian group structure by the following addition operation: Addition formula for char $F \neq 2, 3$: Let $P = (x_1, y_1) \neq O$ be a point, the inverse of P is $-P = (x_1, -y_1)$. Let $Q = (x_2, y_2) \neq O$ be a second point with $Q \neq -P$, the sum $P + Q = (x_3, y_3)$ can be calculated as:

$$\begin{cases} x_3 = \lambda^2 - x_1 - x_2 \\ y_3 = \lambda(x_1 - x_3) - y_1 \end{cases} \quad (4)$$

$$\text{with } \lambda = \begin{cases} \frac{y_2 - y_1}{x_2 - x_1}, & \text{if } P \neq Q, \\ \frac{3x_1^2 + a}{2y_1}, & \text{if } P = Q, \end{cases} \quad (5)$$

The operation of $P + Q$ is called point addition (A) if $P \neq Q$, and it is called point doubling (D) if $P = Q$. Obviously, point addition differs from point doubling in formulas. Figure 2 demonstrates the ECC point doubling and addition operations.

2) Scalar multiplication

The operation of adding a point P to itself k times is called

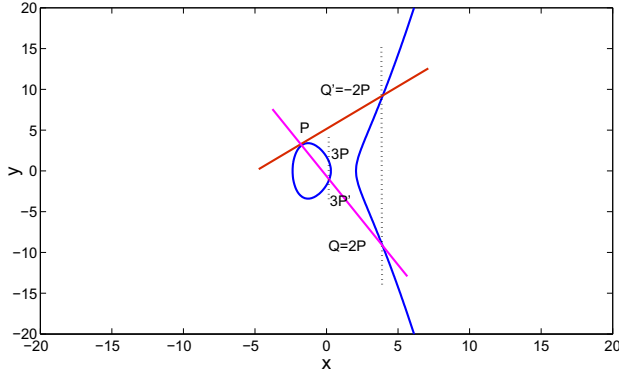


Figure 2. Point addition and point doubling in ECC

scalar multiplication by k and denoted as kP . Scalar multiplication is the basic key-involved operation for ECC, thus the main target in side-channel attacks. The most straightforward implementation of scalar multiplication is the binary algorithm [8] based on the binary expansion, where $n - 1$ is the most significant bit of k . Table 1 illustrates the operation of the scalar multiplication with the binary algorithm (right-to-left).

Table 1. Scalar Multiplication – Binary Algorithm (right-to-left)

Input: $k = (k_{n-1}, \dots, k_1, k_0)_2$ ($k_{n-1} = 1$), $P \in E(Fp)$
Output: kP
1. $Q = \infty$
2. For i from 0 to $n-1$ do
2.1 If $k_i = 1$ then $Q = Q + P$ %addition
2.2 $Q = 2P$ %doubling
3. Return (Q)

Generally, SPA against ECC binary algorithm is to learn the private key k by observing the different power patterns for key bit “1” and key bit “0”. For example, the classic binary algorithm consists of a point doubling operation if the key bit is “0”, and a point doubling followed by a point addition operation if the key bit is “1”. If the side channel pattern of doubling is different from that of addition, attackers can easily retrieve the secret key from a single power trace. Figure 3 demonstrate a different power consumption patterns of doubling and addition operations. However, most current implementations of scalar multiplication use the same atom-operations [9] to implement doubling and addition, it is therefore difficult to distinguish between doubling and addition operations for random input. Figure 4 demonstrates the nearly same power consumption patterns of doubling and addition operations with the atom-

icity countermeasure [9] against SPA.

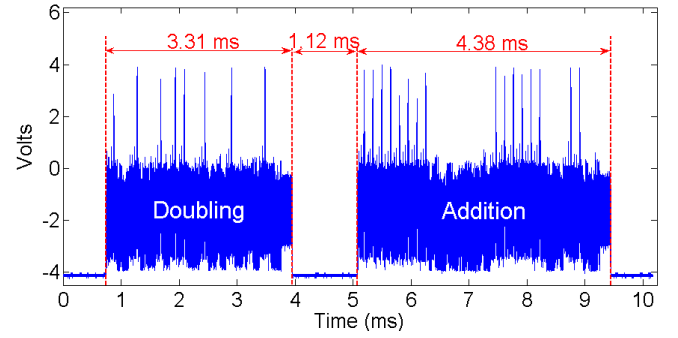


Figure 3. Different power consumption patterns between doubling and addition operations

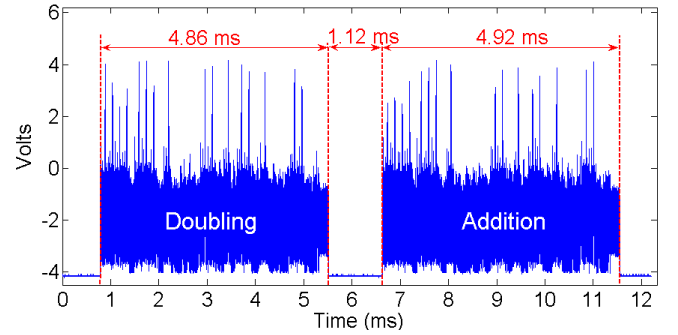
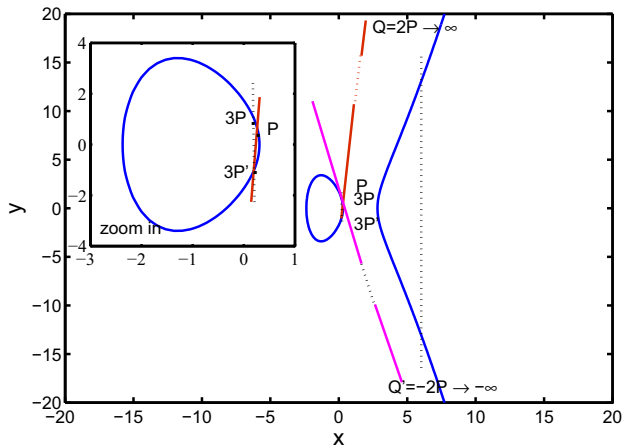


Figure 4. Nearly same power consumption patterns of doubling and addition operations with the atomicity countermeasure

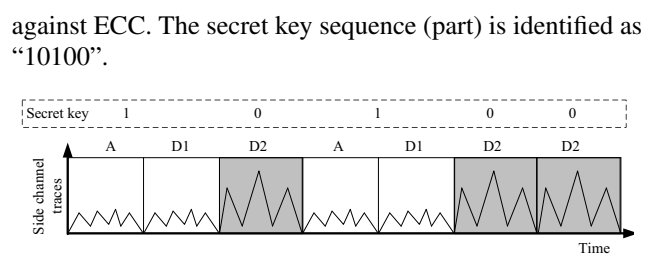
3 Proposed chosen-message SPA against ECC binary algorithm

Since most current implementations of scalar multiplication use the same sequence of instructions to implement point addition and doubling, the differences between addition and doubling in the form of power traces are not always observable through the SPA attack. In order to enhance secret information leakage on the power waveforms, chosen-message attacks for ECC using specific input data is proposed. The basic concept is to use a specific input data, i.e., the point P close to X-axis (denoted as P_x in this paper), to enhance the differences between the doubling and addition operations. Point doubling of such input $Q = 2P_x$ leads to a point near to infinity, i.e., the coordinate values (x_Q, y_Q) are very large, which generates considerable amount of power consumption. Such point Q near to infinity is denoted as Q_∞ in this paper.



According to the binary algorithm (right-to-left) as shown in Table 1, the doubling and addition can be classified into three types: (A) addition, (D1) doubling after addition, (D2) doubling after doubling. According to Eq (4-5), D2 will consume much larger amount of power than D1 or A operations. Thus, even if an implementation of the scalar multiplication has balanced addition and doubling operations with atomicity countermeasures, as shown in Figure 4, our specially chosen input point P_x will make (D2) distinguishable from (D1) and (A). Figure 6 illustrates the difference between D1, D2 and A operations in the form of side channel trace patterns.

Since (D2) only appears when the key bit being processed is “0”, the key bit sequence can be thus identified through SPA. Figure 7 illustrates the chosen-message SPA



4 Extension of the proposed attack

This method magnifies the power consumption amplitude in doubling after doubling (D2) operations with specially chosen input point P_x , and can be extend to left-to-right binary algorithm and the double-and-add-always algorithm.

4.1 Attack on left-to-right binary algorithm

Table 2 illustrates the operation of the scalar multiplication with the binary algorithm (left-to-right).

Input: $k = (k_{n-1}, \dots, k_1, k_0)_2$, $P \in E(Fp)$ ($k_{n-1} = 1$) Output: kP
1. $Q = \infty$ 2. For i from n-1 to 0 do 2.1 $Q = 2P$ %doubling 2.2 If $k_i = 1$ then $Q = Q + P$ %addition 3. Return (Q)

According to the above left-to-right binary algorithm, the doubling and addition can be classified into three types: (A) addition, (D1) doubling after addition, (D2) doubling after doubling, similarly to those in the left-to-right binary algorithm. D2 will consume much larger amount of power than D1 or A operations. Thus the secret key bit sequence can be visually read by an adversary.

4.2 Attack on double-and-add-always algorithm

The double-and-add-always algorithm [8] is a typical SPA countermeasure in ECC implementations. It inserts dummy operations for the left-to-right binary algorithm. Table 3 illustrates the operation of the scalar multiplication

with double-and-add-always algorithm, where the dummy addition is processed for the secret key bit of “0”, in order to perform doubling and addition for each bit.

Table 3. Scalar Multiplication – Double-and-Add-Always Algorithm

Input: $k = (k_{n-1}, \dots, k_1, k_0)_2, P \in E(Fp) (k_{n-1} = 1)$
Output: kP
1. $Q_0 = P$
2. For i from n-2 to 0 do
2.1 $Q_0 = 2Q_0$; %doubling
2.2 $Q_1 = Q_0 + P$; %addition
2.3 $Q_0 = Q_{k_i}$
3. Return (Q_0)

The dummy addition is inserted after D2, denoted as “DummyA” in the following Eq(12). Comparing the dummy addition (DummyA) to the normal addition (A) with the specially chosen input P_x , one can notice the “DummyA” operation will consume smaller amount of power than the “A” operation, making the dummy addition distinguishable. Since the dummy addition will only appear when the key bit is “0”, an adversary can obtain the key sequence.

$$(A)Q_x = Q_\infty + P_x \quad (9)$$

$$(D1)Q_\infty = 2P_x \quad (10)$$

$$(D2)Q_x = 2P_\infty \quad (11)$$

$$(DummyA)Q'_x = Q_x + P_x \quad (12)$$

5 Conclusion

This paper proposed new simple power analysis attacks based on chosen-message input, which can be applied to a variety of scalar multiplication algorithms. The special input point P_x is selected to be close to X-axis, in order to exploit the special characteristics of the point at infinity. This method magnifies the power consumption amplitude in doubling after doubling (D2) operations. The proposed attacks can be adapted to all standard binary algorithms of ECC implementations, namely left-to-right and right-to-left binary algorithms. Typical SPA countermeasures as double-and-add-always algorithm can also be defeated by this new attack method. Further research aiming to expand the scope of the attacks to the Montgomery algorithm and the sliding-window algorithm etc. Practical experiments on real ECC implementations is underway.

Acknowledgement

This work was supported by the National Natural Science Foundation of China (Grant No. 60901052), and the Basic Research Project of Shenzhen (Grant No. JC200903160412A).

References

- [1] V.S. Miller. Use of Elliptic Curves in Cryptography. *Advances in Cryptology - CRYPTO '85: Proceedings*, pages 417–426, 1986.
- [2] N. Koblitz. Elliptic Curve Cryptosystems. *Mathematics of Computation*, 48(177):203–209, 1987.
- [3] P. Kocher, J. Jaffe, and B. Jun. Timing attacks on implementations of Diffie-Hellman, RSA, DSS, and other systems *Proceedings of 16th International Advances in Cryptology Conference – CRYPTO '96*, pages 104–113, 1996.
- [4] A. Miyamoto, N. Homma, T. Aoki, and A. Satoh. Enhanced power analysis attack using chosen message against RSA hardware implementations. pages 3282–3285. IEEE, 2008.
- [5] R. Novak. SPA-based adaptive chosen-ciphertext attack on RSA implementation. In *PKC: International Workshop on Practice and Theory in Public Key Cryptography*. LNCS, 2002.
- [6] B.D. Boer, K. Lemke, and G. Wicke. A DPA attack against the modular reduction within a CRT implementation of RSA. In *CHES: International Workshop on Cryptographic Hardware and Embedded Systems*, CHES, LNCS, 2002.
- [7] S.M. Yen, W.C. Lien, S.J. Moon, and J. Ha. Power analysis by exploiting chosen message and internal collisions - vulnerability of checking mechanism for rsa-decryption. pages 183–195, 2005.
- [8] J.S. Coron. Resistance against differential power analysis for elliptic curve cryptosystems. In *CHES: International Workshop on Cryptographic Hardware and Embedded Systems*, CHES, LNCS, 1999.
- [9] T. Chen, H. Li, K. Wu, and F. Yu. Countermeasure of ecc against side-channel attacks: Balanced point addition and point doubling operation procedure. In *Asia-Pacific Conference on Information Processing 2009*, volume 2, pages 465–469, 2009.